

DSA Day 41: Heap Sort & Radix Sort

Book: Data Structures Through C in Depth - Ch8 §8.15-8.17 · pp 454-471 | Code in Python

Overview

Heap sort sorts in place in $O(n \log n)$. Radix sort sorts integers digit by digit in $O(d \cdot (n+b))$ - non-comparison.

Key concepts to master

- in-place heap sort
- digit buckets
- stable counting per digit

Python implementation

```
def radix(a):
    if not a: return a
    exp=1; mx=max(a)
    while mx//exp>0:
        buckets=[[] for _ in range(10)]
        for x in a: buckets[(x//exp)%10].append(x)
        a=[x for b in buckets for x in b]; exp*=10
    return a
```

Complexity

Heap sort $O(n \log n)$; radix $O(d(n+b))$.

Common pitfalls

- Radix needs non-negative integers here
- Unstable digit pass breaks radix

Practice

- Radix-sort {329,457,657,839}
- When can radix beat $O(n \log n)$?

